FILE: C:\HCL_TRAINING\client\src\pages\ComparisonPage.jsx

LINES: 365 | SIZE: 14.4 KB | LANGUAGE: JSX

MODULE: Side-by-Side Candidate Comparison Matrix for 2 to 4 Applicants

```
1  | import React, { useState, useEffect } from 'react';
2  | import { useSearchParams, Link } from 'react-router-dom';
3  | import {
4  |   GitCompare,
5  |   CheckCircle2,
6  |   XCircle,
7  |   AlertCircle,
8  |   Briefcase,
9  |   GraduationCap,
10 |   Sparkles,
11 |   ArrowLeft,
12 |   ChevronDown,
13 |   Layers,
14 | } from 'lucide-react';
15 | import { api } from '../api/client';
16 | import { ScoreBadge } from '../components/common/ScoreBadge';
17 |
18 | export const ComparisonPage = () => {
19 |   const [searchParams] = useSearchParams();
20 |   const initialJobId = searchParams.get('jobId') || '';
21 |   const initialCandIds = (searchParams.get('candidates') || '').split(',').filter(Boolea
   > n);
22 |
23 |   const [jobs, setJobs] = useState([]);
24 |   const [selectedJob, setSelectedJob] = useState(initialJobId);
25 |   const [availableCandidates, setAvailableCandidates] = useState([]);
26 |   const [selectedCandidates, setSelectedCandidates] = useState(initialCandIds);
27 |   const [comparisonData, setComparisonData] = useState(null);
28 |   const [loading, setLoading] = useState(false);
29 |
30 |   useEffect(() => {
31 |     fetchJobs();
32 |   }, []);
33 |
34 |   useEffect(() => {
35 |     if (selectedJob) {
36 |       fetchJobCandidates(selectedJob);
37 |     }
38 |   }, [selectedJob]);
39 |
40 |   useEffect(() => {
41 |     if (selectedJob && selectedCandidates.length >= 2) {
42 |       runComparison();
43 |     }
44 |   }, [selectedJob, selectedCandidates]);
45 |
46 |   const fetchJobs = async () => {
47 |     try {
48 |       const res = await api.get('/jobs');
49 |       if (res.success && res.data.length > 0) {
50 |         setJobs(res.data);
51 |         if (!selectedJob) setSelectedJob(res.data[0]._id);
52 |       }
53 |     } catch (e) {}
54 |   };
55 |
56 |   const fetchJobCandidates = async (jId) => {
57 |     try {
58 |       const res = await api.get(`/screening/job/${jId}`);
59 |       if (res.success) {
60 |         const cands = (res.data || []).map((r) => r.candidateId).filter(Boolean);
61 |         setAvailableCandidates(cands);
62 |
63 |         // If none selected, default select top 2
64 |         if (selectedCandidates.length < 2 && cands.length >= 2) {
65 |           setSelectedCandidates([cands[0]._id, cands[1]._id]);
```

```
66 |         }
67 |       }
68 |     } catch (e) {}
69 |   };
70 |
71 |   const runComparison = async () => {
72 |     setLoading(true);
73 |     try {
74 |       const res = await api.post('/screening/compare', {
75 |         jobId: selectedJob,
76 |         candidateIds: selectedCandidates.slice(0, 4),
77 |       });
78 |       if (res.success) {
79 |         setComparisonData(res.data);
80 |       }
81 |     } catch (err) {
82 |       console.error(err);
83 |     } finally {
84 |       setLoading(false);
85 |     }
86 |   };
87 |
88 |   const toggleCandidate = (id) => {
89 |     setSelectedCandidates((prev) => {
90 |       if (prev.includes(id)) {
91 |         if (prev.length <= 2) {
92 |           alert('Comparison requires at least 2 candidates.');
93 |           return prev;
94 |         }
95 |         return prev.filter((c) => c !== id);
96 |       } else {
97 |         if (prev.length >= 4) {
98 |           alert('Maximum of 4 candidates can be compared simultaneously.');
99 |           return prev;
100 |         }
101 |         return [...prev, id];
102 |       }
103 |     });
104 |   };
105 |
106 |   const job = comparisonData?.job;
107 |   const cards = comparisonData?.candidates || [];
108 |
109 |   return (
110 |     <div className="space-y-6 max-w-7xl mx-auto">
111 |       {/* Header */}
112 |       <div className="flex flex-col sm:flex-row sm:items-center justify-between gap-4 pb
    >   -2 border-b border-border">
113 |         <div>
114 |           <span className="text-xs font-semibold uppercase tracking-wider text-forest-70
    >   0">
115 |             Side-by-Side Candidate Intelligence
116 |           </span>
117 |           <h1 className="font-editorial text-3xl font-bold text-ink mt-0.5">
118 |             Candidate Comparison Studio
119 |           </h1>
120 |           <p className="text-xs text-ink-muted mt-1">
121 |             Evaluate skill coverage matrices, experience depth, and verified evidence ac
    >   ross 2 to 4 candidates.
122 |           </p>
123 |         </div>
124 |
125 |         {/* Job selector */}
126 |         <div className="flex items-center gap-2">
127 |           <label className="text-xs font-semibold text-ink">Role Context:</label>
128 |           <select
129 |             value={selectedJob}
130 |             onChange={(e) => {
131 |               setSelectedJob(e.target.value);
132 |               setSelectedCandidates([]);
133 |             }}
134 |             className="bg-white border border-border rounded-lg text-xs font-medium text
```

```
  > -ink px-3 py-1.5 focus:outline-none focus:border-forest-500 shadow-subtle min-w-[220px]"
135 |           >
136 |             {jobs.map((j) => (
137 |               <option key={j._id} value={j._id}>
138 |                 {j.title}
139 |               </option>
140 |             ))}
141 |           </select>
142 |         </div>
143 |       </div>
144 |
145 |       {/* Candidate Picker Bar */}
146 |       <div className="surface-card rounded-2xl p-4">
147 |         <span className="text-xs font-semibold text-ink block mb-2">
148 |           Select candidates to compare (2 to 4):
149 |         </span>
150 |         <div className="flex flex-wrap gap-2">
151 |           {availableCandidates.map((cand) => {
152 |             const isSelected = selectedCandidates.includes(cand._id);
153 |             return (
154 |               <button
155 |                 key={cand._id}
156 |                 onClick={() => toggleCandidate(cand._id)}
157 |                 className={`text-xs px-3 py-1.5 rounded-lg border font-medium transition
  > -all ${
158 |                   isSelected
159 |                     ? 'bg-forest-50 border-forest-500 text-forest-800 ring-1 ring-forest
  > -500 shadow-sm'
160 |                     : 'bg-white border-border text-ink-muted hover:border-forest-300'
161 |                 }`}
162 |               >
163 |                 {isSelected ? 'â€“ ' : '+ '}
164 |                 {cand.name}
165 |               </button>
166 |             );
167 |           })}
168 |         </div>
169 |       </div>
170 |
171 |       {/* Comparison Matrix Table */}
172 |       {loading ? (
173 |         <div className="py-16 text-center text-xs text-ink-muted">Running comparative an
  > alysis...</div>
174 |       ) : !comparisonData || cards.length < 2 ? (
175 |         <div className="surface-card rounded-2xl p-12 text-center text-xs text-ink-muted
  > ">
176 |           Please select at least 2 candidates to generate the comparison matrix.
177 |         </div>
178 |       ) : (
179 |         <div className="surface-card rounded-2xl overflow-hidden border border-border sh
  > adow-subtle">
180 |           <div className="overflow-x-auto">
181 |             <table className="w-full text-left border-collapse text-xs">
182 |               <thead>
183 |                 <tr className="bg-[#FAFBF9] border-b border-border">
184 |                   <th className="py-4 px-5 w-48 text-[11px] font-semibold text-ink-muted
  >  uppercase tracking-wider">
185 |                     Criteria Matrix
186 |                   </th>
187 |                   {cards.map((c, i) => (
188 |                     <th key={i} className="py-4 px-5 min-w-[240px] border-l border-borde
  > r align-top">
189 |                       <div className="flex items-center justify-between mb-1">
190 |                         <Link
191 |                           to={`/candidates/${c.candidate?._id}`}
192 |                           className="font-editorial text-base font-bold text-ink hover:t
  > ext-forest-700"
193 |                         >
194 |                           {c.candidate?.name}
195 |                         </Link>
196 |                         <ScoreBadge score={c.screening?.overallScore} size="sm" />
197 |                       </div>
```

```
198 |                       <span className="text-[11px] text-ink-subtle block">
199 |                         {c.candidate?.location || 'Remote'}
200 |                       </span>
201 |                     </th>
202 |                   ))}
203 |                 </tr>
204 |               </thead>
205 |
206 |               <tbody className="divide-y divide-border">
207 |                 {/* 1. Overall Match Score */}
208 |                 <tr className="bg-sage-50/40">
209 |                   <td className="py-3 px-5 font-semibold text-ink text-[11px]">
210 |                     Overall AI Match
211 |                   </td>
212 |                   {cards.map((c, i) => (
213 |                     <td key={i} className="py-3 px-5 border-l border-border">
214 |                       <div className="flex items-center gap-2">
215 |                         <span className="font-mono text-base font-bold text-ink">
216 |                           {c.screening?.overallScore}%
217 |                         </span>
218 |                         <span className="text-[10px] text-ink-muted">weighted score</spa
    >   n>
219 |                       </div>
220 |                     </td>
221 |                   ))}
222 |                 </tr>
223 |
224 |                 {/* 2. Component breakdown */}
225 |                 <tr>
226 |                   <td className="py-3 px-5 font-semibold text-ink text-[11px]">
227 |                     Component Scores
228 |                   </td>
229 |                   {cards.map((c, i) => {
230 |                     const sc = c.screening?.componentScores || {};
231 |                     return (
232 |                       <td key={i} className="py-3 px-5 border-l border-border">
233 |                         <div className="space-y-1 text-[11px] font-mono">
234 |                           <div className="flex justify-between">
235 |                             <span className="text-ink-subtle">Skills (40%):</span>
236 |                             <span className="font-semibold text-ink">{sc.skillAlignment
    >   || 0}%</span>
237 |                           </div>
238 |                           <div className="flex justify-between">
239 |                             <span className="text-ink-subtle">Semantic (30%):</span>
240 |                             <span className="font-semibold text-ink">{sc.semanticSimilar
    >   ity || 0}%</span>
241 |                           </div>
242 |                           <div className="flex justify-between">
243 |                             <span className="text-ink-subtle">Exp (20%):</span>
244 |                             <span className="font-semibold text-ink">{sc.experienceScore
    >    || 0}%</span>
245 |                           </div>
246 |                           <div className="flex justify-between">
247 |                             <span className="text-ink-subtle">Qual (10%):</span>
248 |                             <span className="font-semibold text-ink">{sc.qualificationSc
    >   ore || 0}%</span>
249 |                           </div>
250 |                         </div>
251 |                       </td>
252 |                     );
253 |                   })}
254 |                 </tr>
255 |
256 |                 {/* 3. Mandatory Skills Matrix */}
257 |                 <tr className="bg-[#FAFBF9]">
258 |                   <td colSpan={cards.length + 1} className="py-2.5 px-5 font-semibold up
    >   percase tracking-wider text-[10px] text-forest-800">
259 |                     Mandatory Skill Coverage Checklist
260 |                   </td>
261 |                 </tr>
262 |
263 |                 {(job?.mandatorySkills || []).map((mSkill) => (
```

```
264 |                     <tr key={mSkill.name}>
265 |                       <td className="py-2.5 px-5 font-medium text-ink text-[11px]">
266 |                         {mSkill.name}
267 |                         <span className="text-[10px] text-ink-subtle block">
268 |                           Min {mSkill.minYears} yrs
269 |                         </span>
270 |                       </td>
271 |                       {cards.map((c, i) => {
272 |                         const matched = (c.screening?.matchedRequiredSkills || []).find(
273 |                           (s) => s.skill.toLowerCase() === mSkill.name.toLowerCase()
274 |                         );
275 |
276 |                         return (
277 |                           <td key={i} className="py-2.5 px-5 border-l border-border">
278 |                             {matched ? (
279 |                               <div className="flex items-start gap-1.5 text-forest-800">
280 |                                 <CheckCircle2 className="w-4 h-4 text-forest-600 shrink-0
    >  mt-0.5" />
281 |                                 <div>
282 |                                   <span className="font-semibold text-[11px]">Matched</spa
    >  n>
283 |                                   <span className="text-[10px] text-ink-muted italic block
    >   line-clamp-1">
284 |                                     "{matched.evidence}"
285 |                                   </span>
286 |                                 </div>
287 |                               </div>
288 |                             ) : (
289 |                               <div className="flex items-center gap-1.5 text-amber-800">
290 |                                 <XCircle className="w-4 h-4 text-amber-600 shrink-0" />
291 |                                 <span className="font-semibold text-[11px]">Missing</span>
292 |                               </div>
293 |                             )}
294 |                           </td>
295 |                         );
296 |                       })}
297 |                     </tr>
298 |                   ))}
299 |
300 |                   {/* 4. Total Experience */}
301 |                   <tr>
302 |                     <td className="py-3 px-5 font-semibold text-ink text-[11px]">
303 |                       Total Relevant Experience
304 |                     </td>
305 |                     {cards.map((c, i) => (
306 |                       <td key={i} className="py-3 px-5 border-l border-border font-mono">
307 |                         <span className="font-bold text-ink">
308 |                           {c.candidate?.totalExperienceYears || 0} Years
309 |                         </span>
310 |                         <span className="text-[10px] text-ink-muted block font-sans">
311 |                           (Job requires {job?.minYearsExperience}y)
312 |                         </span>
313 |                       </td>
314 |                     ))}
315 |                   </tr>
316 |
317 |                   {/* 5. Highest Education */}
318 |                   <tr>
319 |                     <td className="py-3 px-5 font-semibold text-ink text-[11px]">
320 |                       Education & Credentials
321 |                     </td>
322 |                     {cards.map((c, i) => {
323 |                       const highestEdu = (c.candidate?.education || [])[0];
324 |                       return (
325 |                         <td key={i} className="py-3 px-5 border-l border-border">
326 |                           <span className="font-semibold text-ink block text-[11px]">
327 |                             {highestEdu?.degree || "Bachelor's Degree"}
328 |                           </span>
329 |                           <span className="text-[10px] text-ink-muted">
330 |                             {highestEdu?.institution || 'Accredited University'}
331 |                           </span>
332 |                         </td>
```

```
                      );
                    })}
                  </tr>

                  {/* 6. Missing Requirements Overview */}
                  <tr className="bg-amber-50/20">
                    <td className="py-3 px-5 font-semibold text-amber-900 text-[11px]">
                      Identified Gaps
                    </td>
                    {cards.map((c, i) => {
                      const missing = c.screening?.missingRequiredSkills || [];
                      return (
                        <td key={i} className="py-3 px-5 border-l border-border text-xs">
                          {missing.length === 0 ? (
                            <span className="text-[11px] text-forest-700 font-medium">None
  >   detected</span>
                          ) : (
                            <span className="text-[11px] text-amber-800 font-medium">
                              Lacks {missing.join(', ')}
                            </span>
                          )}
                        </td>
                      );
                    })}
                  </tr>
                </tbody>
              </table>
            </div>
          </div>
        )}
      </div>
    );
  };

```